Stop Interpolation Topologically

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Agenda

Introduction

Persistent Homology

Triangulation of Manifolds

Persistence Stability

Natural Neighbor Interpolation

The Collapsing Sequence

Hypothesis Testing

Summary

Introduction

Hypothesis

Two main assumptions

Hypothesis 1: Manifold assumption

A point set X lies on a *topological* or even *smooth manifold* $\mathbb{M}$, having a family of continuous/smooth *coordinate systems* to describe it, with $\dim \mathbb{M} \ll \dim X$.

Hypothesis 2: Meaning of shape

The topology of the sublevel sets of a manifold $\mathbb{M}$ underlying a data set can be used to distinguish data up to some equivalence relation.

Persistent Homology

An outline to persistent homology

A brief description

Persistent homology is the study of k-**dimensional abelian groups** attached to a topological space. They are studied algebraically and behave *similar* to the topological space itself under change revealing properties of the original space they are attached to by their *invariants*.

The connection of abelian groups by a boundary homomorphism allows to investigate all k-dimensions at once. The induced object is a chain complex.

Chain complex

How does it work?

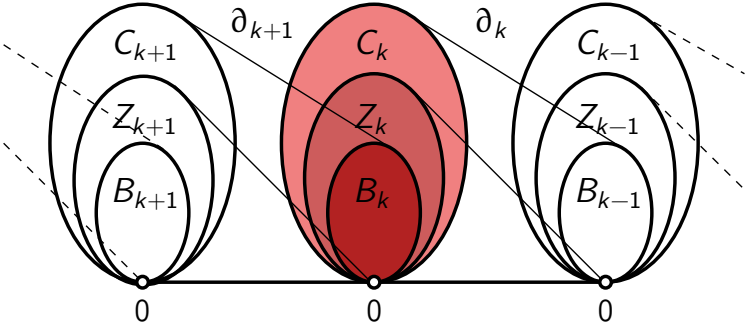
Homology theory is the study of properties of an object X, such as a set of points, by means of homological (commutative) algebra. One assigns to X a sequence of abelian groups/modules $C_0, C_1, \cdots$ which are connected by homomorphisms $\partial_{k+1} : C_{k+1} \longrightarrow C_k$ such that $\mathrm{im}\partial_{k+1} \subseteq \ker \partial_k$. Such a structure is called **chain complex**.

$$0 \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \xrightarrow{\partial_{k-1}} C_{k-2} \xrightarrow{\partial_{k-2}} \cdots \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} 0$$

Cycle and boundary groups

Putting them into context

Illustration of the chain complex



Following Zomorodian, Edelsbrunner and many more ...

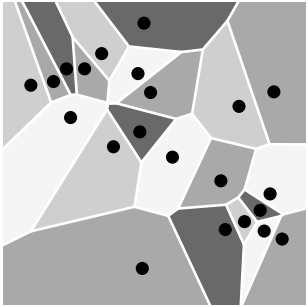
Triangulation of Manifolds

Persistence Stability

Natural Neighbor Interpolation

Step 1: Voronoi diagram

Natural neighbor interpolation example

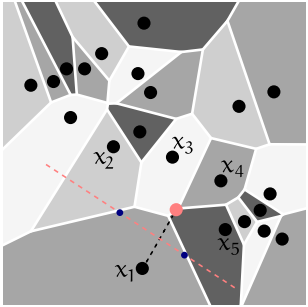


The **Voronoi Diagram** is spanned over the point cloud embedded into some Euclidean space. Technically one does not calculate the Voronoi diagram but the dual **Delaunay triangulation**.

$$\text{Vordgm} = \{x \in X \mid d(x, \lambda_i) \leq d(x, \lambda_j)\} \qquad (1)$$

Step 2: Add a point to the convex hull

We need to compute the weights per region

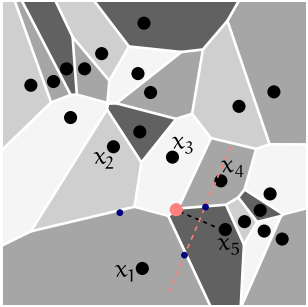


We insert a point into the convex hull of the points $X = \{x_1, \cdots, x_n\}$. The selected position of the new point follows an uniform distribution. The point is added to the set forming the convex hull:

$$\text{conv}X = \left\{ \sum_{i=1}^{n} \lambda_i x_i \;\middle|\; x_i \in X, \sum_{i=1}^{n} \lambda_i = 1, \lambda_i \geq 0 \right\}.$$

Step 3: Iterative construction of Voronoi regions

Construction of the new regions



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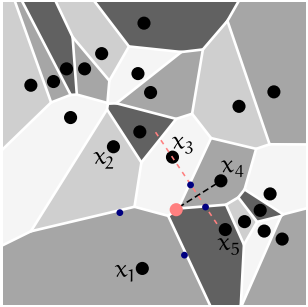

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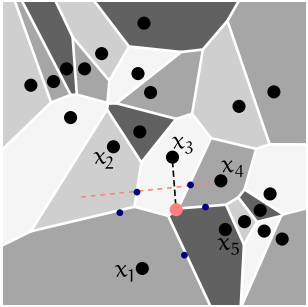

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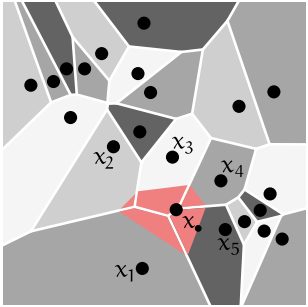
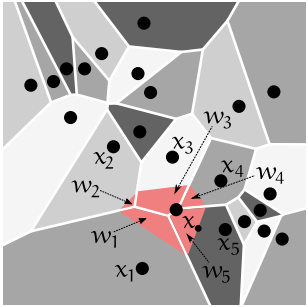


An outline to persistent homology

A brief description



An outline to persistent homology

A brief description

The Collapsing Sequence

Hypothesis Testing

Summary

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